

An upper bound on the Berry–Esseen constant for independent summands

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Abstract

Let Δ be the Kolmogorov distance between a standardized sum of independent centered real random variables and the standard normal law, and let L be the normalized sum of third absolute moments. We propose the uniform bound $\Delta < 0.474999998 L$. The argument treats a summand of maximal variance separately and bounds the real and imaginary parts of the zero-bias error in terms of third-moment excess over symmetric two-point laws. A quadratic term in the logarithm strengthens the characteristic-function product bound. Prawitz’s smoothing inequality then reduces the intermediate range of L to a finite interval calculation; analytic estimates cover small and large L . The supplement contains the arithmetic implementation and complete partition records. Independent mathematical review remains outstanding.

1 The problem and the candidate bound

Let $X_1, \dots, X_n$ be a finite independent family of centered real random variables. Assume

$$0 < V = \sum_{j=1}^n \mathbb{E} X_j^2 < \infty, \quad \sum_{j=1}^n \mathbb{E} |X_j|^3 < \infty.$$

Write Φ and ϕ for the standard normal distribution function and density, and set

$$S = \frac{\sum_j X_j}{\sqrt{V}}, \quad L = \frac{\sum_j \mathbb{E} |X_j|^3}{V^{3/2}}, \quad \Delta = \sup_{x \in \mathbb{R}} |\mathbb{P}(S \leq x) - \Phi(x)|. \quad (1.1)$$

Thus $C_{\text{ind}} = \sup \Delta/L$, over all such arrays, is the unrestricted independent Berry–Esseen constant. Esseen’s lower bound is

$$C_E = \frac{\sqrt{10} + 3}{6\sqrt{2\pi}} = 0.40973218370239634299 \dots \leq C_{\text{ind}}; \quad (1.2)$$

see Esseen [1] and the discussion in Shevtsova [6]. The conjectured equality $C_{\text{ind}} = C_E$ remains open.

Proposed theorem 1.1. *For the arrays in (1.1),*

$$\Delta < 0.474999998 L < 0.475 L, \quad C_E \leq C_{\text{ind}} < 0.474999998. \quad (1.3)$$

The arithmetic assumptions and completed checks for this proposed bound are stated in Section 7. For comparison, Shevtsova [5] obtained the upper bounds 0.5583 for independent summands

and 0.4690 for identically distributed summands. Schulz [4] proved the sharp constant C_E for binomial laws.

The proof uses the zero-bias and Fourier framework of Tyurin [7] and the smoothing inequality recorded in Shevtsova [6]. Comparison with symmetric two-point sums also appears in the smoothing-function results of Mattner and Shevtsova [2]. We derive below the estimates used here, including their aggregation across unequal variances.

1.1 Normalization and moment parameters

Replace each X_j by $X_j/\sqrt{V}$, reuse the notation X_j , and discard summands of variance zero. Then $S = \sum_j X_j$ and the total variance is one. Put

$$v_j = \mathbb{E}X_j^2, \quad b_j = \mathbb{E}|X_j|^3, \quad \tau = \sum_j v_j^{3/2}, \quad \sum_j v_j = 1, \quad \sum_j b_j = L.$$

The differences $b_j - v_j^{3/2}$ are nonnegative and sum to $L - \tau$. They vanish for centered symmetric two-point summands. Relabel the summands so that $v_1 = \max_j v_j$, and put $d = v_1$ and $b = b_1$. The moment parameters satisfy

$$b \geq d^{3/2}, \quad d^{3/2} \leq \tau \leq \min\{L - b + d^{3/2}, \sqrt{d}\}, \quad \tau \leq L. \quad (1.4)$$

Indeed $b_j \geq v_j^{3/2}$ and $v_j \leq d$ imply $\tau \leq \sqrt{d} \sum_j v_j = \sqrt{d}$. In particular $\tau^2 \leq d \leq \tau^{2/3}$. For the remaining summands, write

$$w = 1 - d, \quad B_R = L - b, \quad \tau_R = \tau - d^{3/2}, \quad a_R = \min\{d, w, B_R^{2/3}, \tau_R^{2/3}\}. \quad (1.5)$$

For $j \geq 2$, we have $v_j \leq d$, $v_j \leq w$, $v_j^{3/2} \leq b_j \leq B_R$, and $v_j^{3/2} \leq \tau_R$. Thus $v_j \leq a_R$. Every realizable tuple satisfies these constraints; maximizing over the possibly larger set they define gives an upper bound.

Section 2 bounds products of characteristic functions. Sections 3 and 4 estimate the zero-bias error of each summand and aggregate these errors. Section 5 combines the estimates while treating X_1 separately. Smoothing and the interval calculation occupy Sections 6 and 7.

2 Bounds for characteristic-function products

For a random variable X , let $f_X(t) = \mathbb{E}e^{itX}$. Since $f_X(-t) = \overline{f_X(t)}$, it suffices to bound its modulus for $t \geq 0$. For a nonempty index set J , put

$$V_0 = \sum_{j \in J} v_j, \quad B = \sum_{j \in J} b_j, \quad \tau_0 = \sum_{j \in J} v_j^{3/2},$$

and let a satisfy $v_j \leq a$ for every $j \in J$. We need bounds for the product over J and for the product after any one index is omitted. Products over the empty set equal one.

2.1 Cubic and trigonometric bounds

Define

$$\kappa_* = \sup_{x>0} \frac{\cos x - 1 + x^2/2}{x^3}, \quad \kappa = \frac{49581}{500000} = 0.099162. \quad (2.1)$$

Let θ be the maximizer in (2.1). For $V_0, E_0 > 0$, $t \geq 0$, and $u = E_0 t / V_0$, put

$$\psi(t; V_0, E_0) = \begin{cases} V_0 t^2 / 2 - \kappa_* E_0 t^3, & u < \theta, \\ V_0^3 E_0^{-2} (1 - \cos u), & \theta \leq u \leq 2\pi, \\ 0, & u > 2\pi. \end{cases} \quad (2.2)$$

The characteristic-function bounds in Tyurin [7, Lemma 1] give, with $v = \mathbb{E}X^2 > 0$ and $b = \mathbb{E}|X|^3 \geq v^{3/2}$ in the second formula,

$$\left| \prod_j f_{X_j}(t) \right| \leq \exp\{-\psi(t; V_0, B + \tau_0)\}, \quad (2.3)$$

$$|f_X(t)| \leq A(t; v, b) := \sqrt{1 - 2\psi(t; v, b + v^{3/2})}. \quad (2.4)$$

For fixed parameters, ψ first increases and then decreases with t , and is zero after the cosine branch. On the cubic branch, $\partial_t \psi = t(V_0 - 3\kappa_* E_0 t)$; on the cosine branch, $\partial_t \psi = (V_0^2 / E_0) \sin u \leq 0$. Thus the upper bounds in (2.3)–(2.4) attain their maxima on a frequency interval at an endpoint. On the cosine branch,

$$\begin{aligned} \partial_{V_0} \psi &= \frac{V_0^2}{E_0^2} \{3(1 - \cos u) - u \sin u\} \geq 0, \\ \partial_{E_0} \psi &= \frac{V_0^3}{E_0^3} \{u \sin u - 2(1 - \cos u)\} \leq 0. \end{aligned}$$

The same signs hold on the cubic branch. The maximizing equation for θ makes the formulas agree at the joins. Consequently the upper bounds decrease with the variance argument and increase with the moment argument on the domains just specified.

For $r(x) = (\cos x - 1 + x^2/2)/x^3$, differentiation gives

$$r'(x) = -\frac{h(x)}{2x^4}, \quad h(x) = x^2 + 2x \sin x + 6(\cos x - 1), \quad h'(x) = 4 \cos(x/2) \{x \cos(x/2) - 2 \sin(x/2)\}.$$

The expression in braces decreases from zero on $(0, 2\pi)$. Consequently h decreases on $(0, \pi)$ and increases on $(\pi, 2\pi)$, with $h(\pi) = \pi^2 - 12 < 0$ and $h(2\pi) = 4\pi^2 > 0$. Thus h has exactly one zero in $(\pi, 2\pi)$. For $x \geq 2\pi$, $r(x) \leq 1/(4\pi) < 3/32 \leq r(4)$. The zero is therefore the unique global maximizer of r . The scalar checker isolates it in (3.99, 4) and encloses $\kappa_* < 0.099161913515 < \kappa$ with Arb, a library for rigorous ball arithmetic.

The cubic expression $V_0 t^2 / 2 - \kappa E_0 t^3$ is a lower bound for (2.2) at every frequency. The numerical cosine lower bound is used only when the entire argument interval lies in $[4, 2\pi]$, which is contained in the cosine branch since $\theta < 4$.

2.2 A logarithmic correction

For each summand indexed by J , define

$$h_j = \frac{v_j t^2}{2} - \kappa(b_j + v_j^{3/2})t^3, \quad q_j = \max(h_j, 0), \quad H = \frac{V_0 t^2}{2} - \kappa(B + \tau_0)t^3.$$

Since $b_j \geq v_j^{3/2}$,

$$0 \leq q_j \leq m(t, a) := \max_{0 \leq x \leq t\sqrt{a}} (x^2/2 - 2\kappa x^3) \leq \frac{1}{216\kappa^2} < \frac{1}{2}. \quad (2.5)$$

The maximum in m occurs at $x = \min(t\sqrt{a}, 1/(6\kappa))$. The single-summand bound gives $|f_{X_j}(t)| \leq \sqrt{1-2q_j}$. Applying $-\frac{1}{2}\log(1-2q) \geq q+q^2$ and using $\sum_j q_j \geq \sum_j h_j = H$ gives

$$\prod_j |f_{X_j}(t)| \leq \exp \left\{ -H - \sum_j q_j^2 \right\}. \quad (2.6)$$

Lemma 2.1. *Let $A_0 = t^2/2 - 2\kappa\sqrt{a}t^3$. When $A_0 \geq 0$,*

$$\sum_j q_j^2 \geq Q := \left(\max \left\{ 0, A_0 \frac{\tau_0}{\sqrt{V_0}} - \kappa t^3 (B - \tau_0) \right\} \right)^2. \quad (2.7)$$

For $A_0 < 0$, take $Q = 0$.

Proof. Since $v_j \leq a$ and $q_j \geq h_j$,

$$q_j + \kappa t^3 (b_j - v_j^{3/2}) \geq v_j (t^2/2 - 2\kappa\sqrt{a}t^3) \geq A_0 v_j.$$

When $A_0 \geq 0$, taking ℓ_2 norms and applying the triangle inequality gives

$$\|(q_j)\|_2 \geq A_0 \|(v_j)\|_2 - \kappa t^3 (B - \tau_0).$$

Cauchy–Schwarz gives $\tau_0 \leq \sqrt{V_0} \|(v_j)\|_2$. Taking the maximum of the right side and zero, then squaring, proves (2.7). $\square$

After omitting index i , the linear sum is $H - h_i \geq H - m$. Also $\sum_{j \neq i} q_j^2 \geq \max(0, Q - m^2)$. Hence

$$\begin{aligned} F_0(t; V_0, B, a, \tau_0) &= \min\{1, e^{-H-Q}\}, \\ D_0(t; V_0, B, a, \tau_0) &= \min\{1, e^{-H+m-\max(0, Q-m^2)}\}. \end{aligned} \quad (2.8)$$

Here D_0 bounds the product after any one index is omitted. We may also omit the quadratic correction in either expression. When several upper bounds apply, we take their pointwise minimum. For the empty index set, set $F_0 = D_0 = 1$.

To bound (2.8) when τ_0 ranges over an interval $[\ell, h]$, suppose first that $A_0 \geq 0$ and $t > 0$. Put

$$k = \kappa t^3, \quad a_0 = A_0/\sqrt{V_0} + k, \quad c_0 = kB.$$

Apart from terms independent of τ_0 , either exponent has the form

$$k\tau_0 - \max\{0, (a_0\tau_0 - c_0)_+^2 - m_0^2\}, \quad z_+ = \max(z, 0),$$

where $m_0 = 0$ for F_0 and $m_0 = m$ for D_0 . Its derivative is k until $a_0\tau_0 - c_0 = m_0$, and then $k - 2a_0(a_0\tau_0 - c_0)$. It is maximized on $[\ell, h]$ at

$$\min\{h, \max\{\ell, \tau_*\}\}, \quad \tau_* = \frac{c_0 + \max\{m_0, k/(2a_0)\}}{a_0}.$$

When $A_0 < 0$, $Q = 0$ and the exponent is increasing in τ_0 . At $t = 0$ both product bounds equal one.

3 Real and imaginary zero-bias errors

For a centered variable of positive variance v , its zero-bias law is defined by

$$v\mathbb{E}g'(X^*) = \mathbb{E}[Xg(X)] \quad (3.1)$$

for absolutely continuous test functions for which the expectations exist. The first Wasserstein distance between two laws is the infimum of $\mathbb{E}|U - V|$ over couplings with those laws. By Tyurin [7, Theorem 3], the distance between X and X^* is at most $\mathbb{E}|X|^3/(2v)$. Applying this bound to e^{itx} gives

$$|f_X(t) - f_{X^*}(t)| \leq |t| \frac{\mathbb{E}|X|^3}{2v}. \quad (3.2)$$

To retain information lost by this bound, let W have mean zero and variance one, and put $Y = |W|$, $\rho = \mathbb{E}Y^3$, and $e = \rho - 1$. Then $e \geq 0$ and

$$\mathbb{E}[(Y - 1)^2(Y + 1/2)] = e. \quad (3.3)$$

At $e = 0$, W has the Rademacher distribution: it takes the values -1 and 1 with probability $1/2$ each. Its characteristic function is $\cos x$, and that of its zero-bias law is $\text{sinc } x$.

Lemma 3.1. *For every real x ,*

$$|\Re f_W(x) - \cos x| \leq e|x|^3/6, \quad |\Re f_{W^*}(x) - \text{sinc } x| \leq e|x|/2, \quad (3.4)$$

where $\text{sinc } x = \sin(x)/x$ with its continuous value at zero.

Proof. For an even function f with Lipschitz continuous second derivative, define

$$R_f(y) = f(y) - f(1) - \frac{f'(1)}{2}(y^2 - 1), \quad y \geq 0.$$

Taylor's integral formula gives $R_f(y) = \int_0^{\max(1,y)} K_y(u) f'''(u) du$ with a nonnegative kernel K_y . If $y \leq 1$, the kernel is $u(y-1)^2/2$ on $[0, y]$ and $(1-u)(u-y^2)/2$ on $[y, 1]$. If $y \geq 1$, it is $u(y-1)^2/2$ on $[0, 1]$ and $(y-u)^2/2$ on $[1, y]$. Its integral is $(y-1)^2(y+1/2)/6$. Apply this identity to $f(y) = \cos(xy)$, use $\|f'''\|_\infty \leq |x|^3$, and take expectations in (3.3). The tangent's quadratic term has expectation zero because $\mathbb{E}Y^2 = 1$.

Define the square-biased law by $\mathbb{E}g(W^\square) = \mathbb{E}[W^2g(W)]$ for bounded measurable g . The zero-bias representation is $W^* = UW^\square$, with U uniform on $[0, 1]$ and independent of $W^\square$. For $z \geq 0$,

$$\mathbb{P}(|W^*| \leq z) = \mathbb{E}[Y \min(z, Y)] \leq \min(z, 1).$$

Thus $|W^*|$ stochastically dominates $\text{Unif}[0, 1]$. Their first Wasserstein distance is their mean difference $\rho/2 - 1/2 = e/2$. Apply the Lipschitz bound for $y \mapsto \cos(xy)$. $\square$

Lemma 3.2. *For all real x ,*

$$|\Im(f_W(x) - f_{W^*}(x))| \leq (|x|/2 + |x|^3/6)\sqrt{e(\rho + 5/3)}, \quad (3.5)$$

$$|\Im f_W(x)| \leq |x|^3\sqrt{e(\rho + 5/3)}/6. \quad (3.6)$$

Proof. It suffices to take $x > 0$. Define $A(y) = x \operatorname{sinc}(xy) + \cos(xy)/x$. The identity (3.1) and centering give

$$\Im(f_W(x) - f_{W^*}(x)) = \mathbb{E}[W(A(Y) - A(1))].$$

Since $|\operatorname{sinc}'(z)| \leq |z|/3$, we have $|A'(y)| \leq x(1 + x^2/3)y$ and hence

$$|A(y) - A(1)| \leq (x/2 + x^3/6)|y^2 - 1|.$$

Cauchy–Schwarz and (3.3) give

$$\mathbb{E}[Y|Y^2 - 1|] \leq \sqrt{e \mathbb{E}\left[\frac{Y^2(Y+1)^2}{Y+1/2}\right]} \leq \sqrt{e(\rho + 5/3)}.$$

The final inequality follows from

$$\frac{y^2(y+1)^2}{y+1/2} = y^3 + \frac{3}{2}y^2 + \frac{y^2}{4(y+1/2)}, \quad \frac{y^2}{y+1/2} \leq \frac{2+4y^2}{9}.$$

The second inequality is equivalent to $(y-1)^2(4y+1) \geq 0$. For (3.6), repeat the same argument with $A(y) = x \operatorname{sinc}(xy)$ and $|A'(y)| \leq x^3 y/3$. $\square$

4 A quadratic minorant and the aggregate error

Lemma 4.1. *For $0 < x \leq 2$, put*

$$F_x(y) = \cos(xy) - y \sin(xy)/x, \quad A_x = \cos x + (1 + x^2) \operatorname{sinc} x.$$

Then $F_x(y) \geq F_x(1) - A_x(y^2 - 1)/2$ for $y \geq 0$. Consequently

$$\Re(f_W(x) - f_{W^*}(x)) \geq \cos x - \operatorname{sinc} x. \quad (4.1)$$

Proof. Set $h_x(z) = \cos z + (1 + x^2) \operatorname{sinc} z$ and $A_x = h_x(x)$. The tangent is $F_x(1) - A_x(y^2 - 1)/2$. Multiply the difference from this tangent by x^2 and put $z = xy$. Its derivative is $z[A_x - h_x(z)]$.

The function h_x is strictly decreasing on $(0, \pi)$. On $[\pi, 2\pi]$ it is at most one, and for $z \geq 2\pi$ it is at most $1 + 5/(2\pi) < 11/6$. Alternating Taylor lower bounds give

$$A_x \geq p_0(x^2), \quad p_0(u) = 2 + u/3 - 7u^2/60 + 17u^3/2520 - u^4/5040.$$

For $0 \leq u \leq 4$, $p_0''(u) = (-98 + 17u - u^2)/420 < 0$, so $p_0(u) \geq \min(p_0(0), p_0(4)) = 194/105 > 11/6$. The tangent difference decreases up to $z = x$ and increases afterwards. Its minimum is zero. Taking expectations, and using $\mathbb{E}Y^2 = 1$, proves (4.1). $\square$

For $0 \leq x \leq 2$, let $g = \operatorname{sinc} x - \cos x \geq 0$ and $c = x/2 + x^3/6$. Lemmas 3.1 and 4.1 give

$$-g \leq \Re(f_W - f_{W^*}) \leq -g + ce, \quad |\Re(f_W - f_{W^*})|^2 \leq g^2 + c^2 e^2. \quad (4.2)$$

The alternating series gives the increasing upper bound

$$\frac{2g}{x} \leq \alpha(x) := \begin{cases} 2x/3 - x^3/15 + x^5/420, & 0 \leq x \leq 2, \\ 2x/3, & x > 2. \end{cases} \quad (4.3)$$

The quotient at zero is defined by continuity. On $[0, 2]$ the polynomial is at most $2x/3$. Put $q(x) = 1 + x^2/3$. Combining (4.2) with (3.5) and (3.2), we obtain

$$|f_W(x) - f_{W^*}(x)| \leq \frac{|x|}{2} \min \left\{ \rho, \sqrt{\alpha(|x|)^2 + q(x)^2(\rho - 1)(2\rho + 2/3)} \right\}. \quad (4.4)$$

For $x > 2$ the square root is at least ρ , so the minimum simply uses (3.2). Reflection and continuity handle negative x and zero.

Proposition 4.2. *For a nonempty subset of summands with moment sums B, τ_0 and individual variances at most a ,*

$$\sum_j v_j |f_{X_j}(t) - f_{X_j^*}(t)| \leq \frac{t}{2} P(t; B, a, \tau_0), \quad t \geq 0, \quad (4.5)$$

where, with $x = t\sqrt{a}$,

$$P(t; B, a, \tau_0) = \min \left\{ B, \sqrt{\alpha(x)^2 \tau_0^2 + q(x)^2 (B - \tau_0)(2B + 2\tau_0/3)} \right\}. \quad (4.6)$$

The bound P increases with t, a, B and decreases with τ_0 . For the empty subset, set $P = 0$.

Proof. The square root in (4.4) is increasing in x and concave in $\rho \geq 1$. In the variable $e = \rho - 1$, its quadratic has discriminant $(64/9)q^4 - 8q^2\alpha^2 > 0$, since $\alpha \leq 2x/3$ and $q^2 - 4x^2/3 = (1 - x^2/3)^2 \geq 0$. The second derivative of the square root has the opposite sign to this discriminant; continuity handles a zero radicand at an endpoint. After scaling (4.4) to variance v_j , each term has the factor $tv_j^{3/2}/2$ and moment ratio $\rho_j = b_j/v_j^{3/2}$. Replace $t\sqrt{v_j}$ by $t\sqrt{a}$ and apply Jensen's inequality with weights $v_j^{3/2}/\tau_0$; their weighted mean of ρ_j is B/τ_0 . This gives the square root in (4.6). The classical bounds also give $tB/2$.

The expression under the square root in (4.6) is

$$2q^2B^2 - \frac{4}{3}q^2B\tau_0 + (\alpha^2 - 2q^2/3)\tau_0^2.$$

Its nonconstant coefficients in τ_0 are nonpositive. Its other monotonicities follow from the original nonnegative-factor expression. $\square$

4.1 A characteristic-function bound using moment excess

Squaring the real-part estimate in Lemma 3.1 and adding the square of the bound in (3.6) gives another single-summand bound. If $\tau_j = v^{3/2}$ and $C = |\cos(t\sqrt{v})|$, then

$$|f_X(t)|^2 \leq C^2 + \frac{C(b - \tau_j)|t|^3}{3} + \frac{t^6(b - \tau_j)(2b + 2\tau_j/3)}{36}. \quad (4.7)$$

This holds for every real t . We use the smaller of its square root and (2.4), evaluated at $|t|$. The displayed terms increase with $C, b, |t|$ and decrease with τ_j when $b \geq \tau_j$. On a parameter interval, first bound C over the entire cosine argument interval, then apply these monotonicities.

5 A summand of maximal variance

The independent-sum zero-bias identity, with total variance one, is

$$f_{S^*}(t) = \sum_i v_i f_{X_i^*}(t) \prod_{j \neq i} f_{X_j}(t). \quad (5.1)$$

For the remaining summands, use the product bound

$$R(t) = \min\{e^{-\psi(t; w, B_R + \tau_R)}, F_0(t; w, B_R, a_R, \tau_R)\}.$$

If the remainder is empty, set $R(t) = D_R(t) = 1$. Let $A(t)$ be the minimum of the single-summand bounds (2.4) and the square root of (4.7) for X_1 . Let $D_R(t)$ be the bound D_0 for the remaining summands, and $D_S(t)$ the corresponding bound for the whole sum. We also evaluate these bounds with Q omitted and take the smaller interval upper bounds. Proposition 4.2 gives

$$|f_S(t) - f_{S^*}(t)| \leq \frac{t}{2} \{P(t; b, d, d^{3/2})R(t) + P(t; B_R, a_R, \tau_R)A(t)D_R(t)\}. \quad (5.2)$$

The whole-sum estimate is $(t/2)P(t; L, d, \tau)D_S(t)$, and the classical estimate is $tL/2$. Thus $|f_S(t) - f_{S^*}(t)| \leq (tL/2)\mathcal{B}(t)$, where

$$\mathcal{B}(t) = \min \left\{ 1, \frac{P(t; L, d, \tau)D_S(t)}{L}, \frac{P(t; b, d, d^{3/2})R(t) + P(t; B_R, a_R, \tau_R)A(t)D_R(t)}{L} \right\}. \quad (5.3)$$

This formula also defines the endpoint value $\mathcal{B}(0)$.

For the characteristic function itself, use

$$F_S(t) = \min\{e^{-\psi(t; 1, L + \tau)}, A(t)R(t), F_0(t; 1, L, d, \tau)\}. \quad (5.4)$$

All terms use the same actual τ ; an interval evaluation allows its whole range in the parameter box. The calculation also bounds the last term of (5.3) by separating the moment fraction $\lambda = b/L$ from two normalized errors. Put

$$U_1 = \frac{P(t; b, d, d^{3/2})R(t)}{b}, \quad U_R = \frac{P(t; B_R, a_R, \tau_R)A(t)D_R(t)}{B_R}.$$

If $B_R = 0$, set $U_R = 0$. Both quantities lie in $[0, 1]$, and the last term in (5.3) equals $\lambda U_1 + (1 - \lambda)U_R$. Suppose a box gives $\lambda \in [\lambda_-, \lambda_+] \subseteq [0, 1]$ and $U_i \leq \bar{U}_i$. Then

$$\lambda U_1 + (1 - \lambda)U_R \leq \max_{\lambda \in \{\lambda_-, \lambda_+\}} \{\lambda \bar{U}_1 + (1 - \lambda)\bar{U}_R\}. \quad (5.5)$$

The right side is the maximum of an affine function on an interval. We take the smaller of this bound and a direct interval evaluation of the numerator in (5.3), divided by a lower bound for L . When a denominator in the separate bounds for U_i has lower endpoint zero, the bound $U_i \leq 1$ is used.

Write $D(t) = f_S(t) - e^{-t^2/2}$. The zero-bias identity gives $f'_S(t) = -t f_{S^*}(t)$, and hence

$$D'(t) + tD(t) = t(f_S(t) - f_{S^*}(t)), \quad D(0) = 0.$$

Multiplication by $e^{t^2/2}$ and integration yield

$$D(t) = e^{-t^2/2} \int_0^t u(f_S(u) - f_{S^*}(u))e^{u^2/2} du.$$

Taking absolute values gives

$$\frac{|f_S(t) - e^{-t^2/2}|}{L} \leq R_\Delta(t) := e^{-t^2/2} \int_0^t \frac{u^2}{2} \mathcal{B}(u) e^{u^2/2} du. \quad (5.6)$$

6 Smoothing and numerical integration

For $0 < u < 1$, put

$$K(u) = \frac{1-u}{2} + \frac{i}{2} \left((1-u) \cot(\pi u) + \frac{1}{\pi} \right).$$

Prawitz's inequality, in the form stated by Shevtsova [6, Lemma 4.1] and Tyurin [7, eq. (46)], gives for any $T > 0$ and $0 < s < 1$

$$\frac{\Delta}{L} \leq 2 \int_0^s |K(u)| R_\Delta(Tu) du + \frac{2 \int_s^1 |K(u)| F_S(Tu) du + G(T, s)}{L}, \quad (6.1)$$

where

$$G(T, s) = 2 \int_0^s \left| K(u) - \frac{i}{2\pi u} \right| e^{-T^2 u^2/2} du + \frac{E_1(T^2 s^2/2)}{2\pi}, \quad E_1(z) = \int_z^\infty \frac{e^{-v}}{v} dv. \quad (6.2)$$

6.1 Integration over a frequency interval

On a frequency interval $[a, b]$, let $\mathcal{B}_j$ be an upper bound for $\mathcal{B}$, and write

$$I_a = \int_0^a u^2 \mathcal{B}(u) e^{u^2/2} / 2 du.$$

Let A_j be an upper bound for $2\pi u |K(u)|$ on $u \in [a/T, b/T]$. Under the change of variable $t = Tu$, the first term of (6.1), restricted to $[a, b]$, is at most

$$\frac{A_j}{\pi} \left\{ I_a \frac{E_1(a^2/2) - E_1(b^2/2)}{2} + \mathcal{B}_j \left(\frac{b^3 - a^3}{18} - \frac{a^3}{6} \log \frac{b}{a} \right) \right\}. \quad (6.3)$$

To see this, split the integral defining $R_\Delta(t)$ at a . The part up to a contributes $\int_a^b I_a e^{-t^2/2} / t dt$, which equals the first term in braces. For the remaining part, use $e^{-(t^2-u^2)/2} \leq 1$ and interchange the nonnegative integrals:

$$\int_a^b \frac{1}{t} \int_a^t \frac{u^2}{2} du dt = \frac{b^3 - a^3}{18} - \frac{a^3}{6} \log \frac{b}{a}.$$

At $a = 0$, $I_a = 0$ and the expression in braces in (6.3) is $\mathcal{B}_j b^3/18$. In the calculation, I_a is replaced by an upper bound accumulated from the preceding intervals. For $a > 0$, we also integrate the bound $|f_S - e^{-t^2/2}| \leq F_S + e^{-t^2/2}$ over $[a, b]$ and use the smaller of the two upper bounds for that integral.

6.2 The Gaussian correction

Direct subtraction of the singular part of K gives

$$2 \left| K(u) - \frac{i}{2\pi u} \right| = (1-u) \sqrt{1 + h(u)^2}, \quad h(u) = \frac{1}{\pi u} - \cot(\pi u). \quad (6.4)$$

Set $h(0) = 0$ by continuity. The partial-fraction identity [3, eq. 4.22.3] is

$$h(u) = \frac{2u}{\pi} \sum_{n=1}^{\infty} \frac{1}{n^2 - u^2}.$$

For $0 \leq u \leq s < 1$ and any integer $M \geq 1$, define

$$c_M(s) = \frac{\pi^2/6 - \sum_{n=1}^M n^{-2}}{1 - s^2/(M+1)^2}, \quad h_M(u) = \frac{2u}{\pi} \left(\sum_{n=1}^M \frac{1}{n^2 - u^2} + c_M(s) \right).$$

For $n \geq M+1$ and $0 \leq u \leq s$, $1/(n^2 - u^2) \leq n^{-2}/(1 - s^2/(M+1)^2)$. Summing gives $0 \leq h(u) \leq h_M(u)$. In the computation, $c_M(s)$ is replaced by an upper dyadic endpoint. Since $s < 1$, every denominator in the finite sum is positive on $[0, s]$. Hence

$$G(T, s) \leq \int_0^s (1-u) \sqrt{1 + h_M(u)^2} e^{-T^2 u^2/2} du + \frac{E_1(T^2 s^2/2)}{2\pi}. \quad (6.5)$$

The calculation uses $M = 8$ and Arb to enclose the integral in (6.5); the integration routine is described in Section 7.4. We also evaluate the following bound and use the smaller upper endpoint:

$$G(T, s) \leq \int_0^s (1-u + \pi^2 u^2/18) e^{-T^2 u^2/2} du + \frac{E_1(T^2 s^2/2)}{2\pi}, \quad (6.6)$$

also supplied by Shevtsova [6, Lemma 4.1]. Its integral is expressible using the error function and elementary functions.

7 The interval calculation

The interval calculation covers $L \in [L_-, L_+]$, where $L_- \approx 0.006$ and $L_+ \approx 1.15$ are the following exact dyadic rationals, represented by the stored binary64 endpoints:

$$L_- = \frac{3458764513820541}{576460752303423488}, \quad L_+ = \frac{2589569785738035}{2251799813685248}. \quad (7.1)$$

7.1 Small and large moments

For $0 < L \leq L_-$, Corollary 4.18 of Shevtsova [6] gives

$$\Delta/L \leq C_E + 0.3413L^{1/3} \leq C_E + 0.3413L_-^{1/3} < 0.471750509536. \quad (7.2)$$

This corollary applies because $L_- < 0.01$. The scalar checker recomputes the last enclosure.

For every mean-zero variance-one law, Cantelli's inequality and reflection give

$$\Delta \leq \sup_{x>0} \left(\Phi(x) - \frac{x^2}{1+x^2} \right) < 0.540936541549. \quad (7.3)$$

For completeness, at $x \geq 0$ the lower CDF bound follows from Cantelli, while the opposite difference is at most $1 - \Phi(x) \leq 1/2$. Reflection handles $x < 0$, including either one-sided convention at atoms. The displayed supremum exceeds $1/2$. Its stationary point is unique: the logarithmic derivative of $2x/((1+x^2)^2 \phi(x))$ is $(x^2 - 1)^2/(x(1+x^2)) \geq 0$, and the ratio increases from zero to infinity. Arb isolation of the stationary point gives

$$\Delta/L < 0.470379601347 \quad (L \geq L_+). \quad (7.4)$$

7.2 Subdivision of the parameter domain

For each closed interval $L \in [\ell, H]$, the calculation starts from

$$0 \leq d \leq \min\{1, H^{2/3}\}, \quad 0 \leq b \leq H, \quad 0 \leq \tau \leq \min\{1, H\},$$

with outward-rounded endpoints. The constraints in (1.4) tighten this box. A box is accepted when its computed upper bound is finite and below 0.475 throughout the box. It is discarded when the constraints prove it infeasible; otherwise, it is divided into two closed boxes whose union contains it. The stored tree records each split and terminal decision. The finer claimed constant 0.474999998 is checked against the recorded maxima after this subdivision is complete.

For each box, the evaluator tries three pairs of smoothing parameters T, s and takes the minimum of the resulting bounds. Each pair divides $[0, Ts]$ and $[Ts, T]$ into 512 equal frequency intervals. The pairs are chosen by the rules in `verify/kernel/refined_integral.py`, with scale and shift pairs $(1, 0)$, $(1.02, -0.02)$, and $(1.04, -0.02)$ from `verify/kernel/batch_weights.py`. The upper endpoint of the box's τ interval selects an input to these rules through `verify/kernel/gaussian_batch.py`. The resulting binary64 values of T, s are treated as exact dyadic parameters; each bound is evaluated over the actual moment box.

Before subdivision, the calculation tries two bounds uniform over (d, b, τ) . The first uses (7.3) in the looser form $\Delta/L \leq 0.54093655/\ell$. The second applies the smoothing integral with $d_U = H^{2/3}$ and $E_U = 2H$, rounded upward, using

$$\mathcal{B}_U(t) = \min\{1, e^{-t^2/2 + \kappa E_U t^3 + m(t, d_U)}\}, \quad F_U(t) = e^{-\psi(t; 1, E_U)}.$$

These follow from $\tau \leq L \leq H$ and the product bounds of Section 2. If either proves the acceptance bound for the whole L interval, the record has a single accepted node.

Quantity	Value
Closed L intervals	767
Partition nodes	5,577,029
Accepted leaves	2,750,409
Infeasible leaves	38,489
Largest recorded upper endpoint	0.47499999772796003
Upper bound for $L \leq L_-$	< 0.471750509536
Upper bound for $L \geq L_+$	< 0.470379601347

Of the 767 intervals, 545 used the moment-parameter bounds, 215 used the uniform smoothing bound, and seven used the variance bound. Every accepted leaf has a finite upper endpoint. Together with the two complementary estimates, their maximum gives

$$C_{\text{ind}} \leq \max\{0.47499999772796003, 0.471750509536, 0.470379601347\} < 0.474999998.$$

7.3 Completed computation and subsequent checks

The original calculation evaluated every accepted-leaf bound. The supplement records three kinds of subsequent checks:

1. *Saved-record checking* checks file and source-code digests, tree syntax and counts, contiguous L intervals, recorded upper endpoints, and the scalar enclosures in (7.2)–(7.4). It does not recompute the quadrature at accepted leaves.

2. *Geometric replay* uses the archived implementation to reconstruct every box reduction and split, check every infeasibility decision, and verify coverage of all 767 intervals. It also checks the analytic complements.
3. *Numerical replay* additionally recomputes quadrature at accepted leaves. The recorded replay recomputed all 4,944 accepted leaves in three intervals, selected by lower endpoints nearest 0.006, 0.25, and 0.48. The checksums of their ordered numerical results and their maxima matched the original records. Quadrature in the remaining intervals has not all been recomputed.

The records contain these checksums and maxima; individual numerical values can be recovered by repeating the computation. Independent mathematical review remains outstanding.

7.4 Arithmetic assumptions and reproduction

The transcendental constants are enclosed with 100-bit Arb arithmetic. The vector calculation assumes IEEE-754 binary64 elementary operations and square root, rounding to nearest, gradual underflow, and no unsafe fast-math transformations. Interval operations use `nextafter` to round endpoints outward. Nonnegative cumulative sums and dot products are multiplied by $1 + 4N2^{-53}$ and rounded upward, where $N = 512$ is the number of frequency intervals in each range for each smoothing choice. The full factor is used for every prefix of a cumulative sum.

For a nonnegative lower bound p on an exponent, the negative exponential is bounded using

$$P_{16}(x) = \sum_{j=0}^{16} \frac{x^j}{j!}, \quad e^{-p} \leq P_{16}(p/64)^{-64} \quad (0 \leq p < 25).$$

The implementation evaluates this polynomial with downward rounding, takes the reciprocal upward, and performs six upward-rounded squarings. For $p \geq 25$, it uses an upper enclosure of e^{-25} . The routine is `exp_upper_from_nonnegative_lower` in `verify/kernel/interval_bounds.py`.

The Gaussian integral in (6.5) is evaluated by `gaussian_G` in `verify/kernel/gaussian_batch.py`, using Arb’s `acb.integral`. Its callback passes the integration routine’s analyticity requirement to the square root through `sqrt(analytic=analytic)`. The returned ball must be finite and its imaginary part must contain zero. The validity of the integration also depends on Arb’s integration contract and the callback’s branch handling. The summation allowance, exponential bound, and integration procedure are included in the outstanding arithmetic audit.

The nine arithmetic source files are identified by SHA-256 digests. The recorded numerical run-time is Python 3.12.10, NumPy 2.4.6, python-flint 0.8.0, and SciPy 1.15.3. The complete interval records are in `certificates/cover.json`; the proposed bound and scalar endpoints are summarized in `result.json`. The supplement’s `verify/README.md` gives the commands for checking the saved records, reconstructing the partition, and recomputing selected numerical bounds.

8 Possible refinements

Two relaxations remain in the estimates: the triangle inequality replaces the modulus of a sum of complex zero-bias errors by the sum of their moduli, and the moment constraints may admit unrealizable tuples. A finer parameter partition can reduce interval-enclosure error, but cannot recover the information lost in either relaxation.

8.1 Higher-order terms in the logarithm

The following higher-order bounds were not used in the completed calculation. With the notation of Lemma 2.1, Hölder interpolation gives, for every integer $p \geq 2$,

$$\|(v_j)\|_{3/2} \leq \|(v_j)\|_1^{(2p-3)/(3p-3)} \|(v_j)\|_p^{p/(3p-3)}.$$

Since $\|(v_j)\|_{3/2} = \tau_0^{2/3}$ and $\|(v_j)\|_1 = V_0$,

$$\|(v_j)\|_p \geq \frac{\tau_0^{2-2/p}}{V_0^{2-3/p}}.$$

If $A_0 \geq 0$, the same norm triangle inequality gives

$$\sum_j q_j^p \geq Q_p := \left(\max \left\{ 0, A_0 \frac{\tau_0^{2-2/p}}{V_0^{2-3/p}} - \kappa t^3 (B - \tau_0) \right\} \right)^p. \quad (8.1)$$

Take $Q_p = 0$ otherwise. For $0 \leq q < 1/2$,

$$-\frac{1}{2} \log(1 - 2q) = \sum_{p=1}^{\infty} \frac{2^{p-1}}{p} q^p.$$

Truncating the series and using (8.1) gives the full-product bound below. Omitting an index reduces each p th-power sum by at most m^p , which gives the second bound. For any integer $P \geq 2$,

$$\begin{aligned} \prod_j |f_{X_j}(t)| &\leq \exp \left\{ -H - \sum_{p=2}^P \frac{2^{p-1}}{p} Q_p \right\}, \\ \prod_{j \neq i} |f_{X_j}(t)| &\leq \exp \left\{ -H + m - \sum_{p=2}^P \frac{2^{p-1}}{p} \max(0, Q_p - m^p) \right\}. \end{aligned} \quad (8.2)$$

Take the minimum of each displayed bound, one, and any other applicable product bound.

8.2 Aggregation and phase information

The real-part interval in (4.2) gives

$$|\Re(f_W - f_{W^*})|^2 \leq \max\{g^2, (ce - g)^2\}.$$

This improves the single-summand bound $g^2 + c^2 e^2$. Aggregating the sharper expression by Jensen's inequality requires a concavity argument for the resulting function of the moments. Such an argument is not supplied here. Another possibility is to retain relative Fourier phases for several summands while bounding the remaining errors uniformly.

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